LECTURE PRESENTATION

System Analysis & Design

System Interface Design, Navigation Design, Input Design and Output Design

Undergraduate Computer Science

Press → to begin · F for fullscreen

INTRODUCTION

Learning Overview

Six parts — from the principles behind good interfaces to applying them in a complete real-world system.

01 **UI Design Principles**
Foundations of usable, aesthetic, and low-effort interfaces

02 **UI Design Process**
From use scenarios to iterative prototyping and evaluation

03 **Navigation Design**
Principles, controls, components, menus, and messaging

04 **Input Design & Validation**
Source data capture, controls, and multi-tier validation

05 **Output Design & Case Study**
Reports, dashboards, and the end-to-end Tune Source journey

06 **Summary & Discussion**
Key takeaways, seminar activity, and academic discussion

Introduction to UI Design

The user interface connects human users to underlying software capabilities, shaping interaction, satisfaction, and operational efficiency.

Purpose of UI Design

Serves as the vital bridge between human users and underlying software functionality — translating intent into action and system state into clear comprehension.

User Interaction

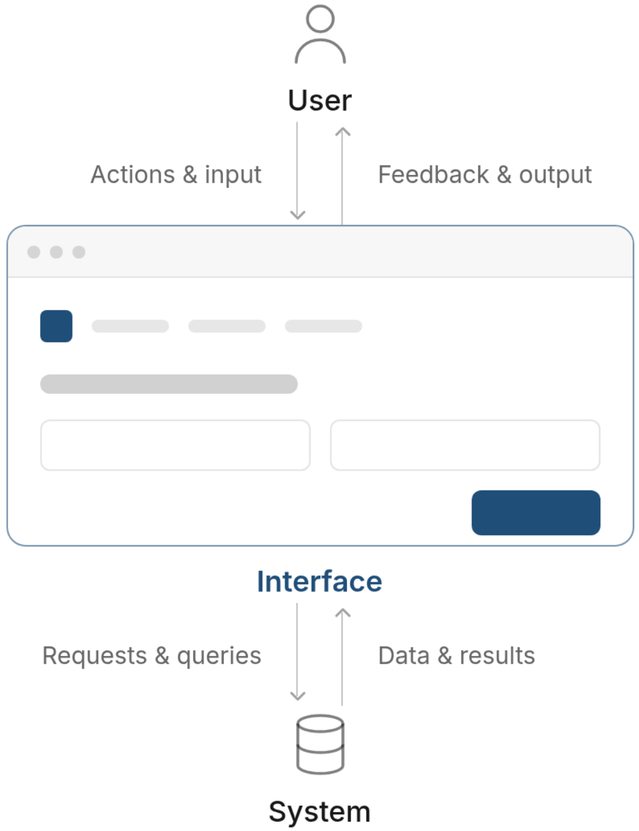
The bidirectional dialogue: users express commands through inputs, gestures, and controls, while the system responds with immediate visual feedback and state changes.

User Satisfaction

How people feel while working with the system — fostering confidence, trust, comfort, and minimal frustration through intuitive, pleasant, and accessible design.

System Efficiency

Enabling tasks to be completed rapidly, accurately, and with minimal friction — eliminating wasted steps, reducing input errors, and maximizing operational throughput.



UI Design Principles

Eight foundational principles that govern the usability, learnability, and longevity of system interfaces.

01

Layout

Organise every screen into predictable, consistent areas (nav, work, status).

02

Content Awareness

Ensure users always know where they are, what they see, and what happened.

03

Aesthetics

Pleasing, functional visual design with balanced whitespace and clear hierarchy.

04

User Experience

Intuitive, satisfying, and easy to learn for novices while efficient for experts.

05

Consistency

Identical behaviors, terms, icons, and visual elements across the system.

06

Minimize User Effort

Reduce cognitive load, keystrokes, and clicks (the Three-Clicks Rule).

07

Accessibility

Ensure equal access across abilities, assistive technologies, and devices.

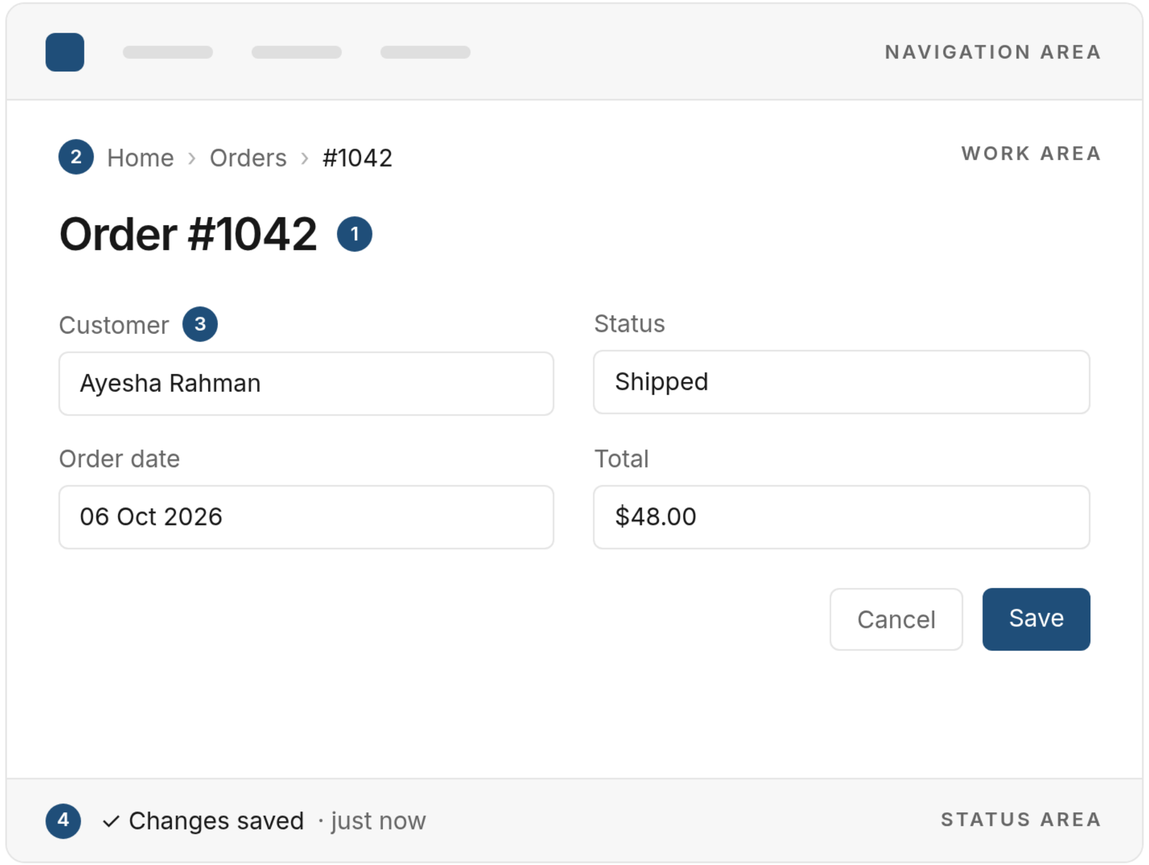
08

Scalability

Accommodate feature expansion, large datasets, and evolving user requirements.

Layout & Content Awareness

Standardized screen zones paired with explicit contextual signposts keep users oriented.



Layout

Divide each screen into three functional zones — navigation, work, and status — and keep their positions uniform across every screen in the system.

Content awareness

Users should never feel lost or uncertain about system state.

1. **Page titles** — Explicitly name the current screen and its purpose
2. **Breadcrumbs** — Show the hierarchical path — and provide a one-click way back
3. **Labels** — Unambiguously describe every input field and content section
4. **Feedback** — Immediately confirm system actions and status changes

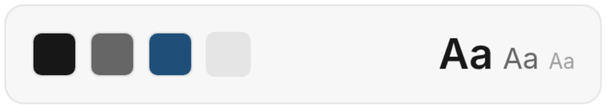
Aesthetics, UX & Consistency

Visual elegance, operational experience, and system-wide predictability work together to eliminate friction.

VISUAL DESIGN — LOOKS

Aesthetics

How the interface appears visually.



Purposeful Colors

A restrained, high-contrast palette. Color reinforces meaning but is never the sole indicator.

Typography Scale

Clear hierarchy across titles, body, and labels with readable, scalable fonts.

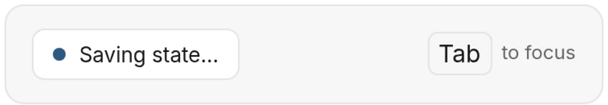
Whitespace & Density

Chunk related elements; give content room to breathe to avoid visual fatigue.

INTERACTION DESIGN — FEELS

User Experience

How the interface feels during operation.



Responsiveness

Immediate visual feedback for every user action; layout adapts seamlessly across devices.

Learnability & Ease

Natural mental models enable novices to learn quickly and experts to work rapidly.

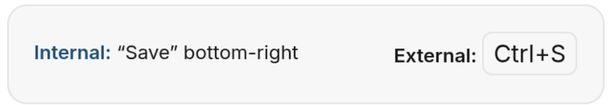
Accessibility & Safety

Full keyboard navigation, high contrast, and tolerant workflows that simplify error recovery.

SYSTEM STANDARDS — BEHAVES

Consistency

How the interface behaves predictably.



Internal Consistency

Same terminology, typography, colors, and button placement on every screen across the system.

External Conventions

Conforms to platform standards users already know (e.g., logo links home, standard shortcuts).

Muscle Memory

Users transfer learned habits across screens without hesitating or relearning interactions.

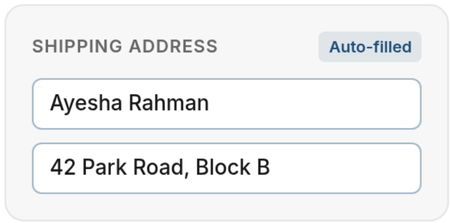
Minimize User Effort

Anticipate user intent, eliminate unnecessary keystrokes, and enforce the Three-Clicks Rule.

EFFORT REDUCTION 01

Autofill

Pre-populate fields using data already stored in profile or browser cache.

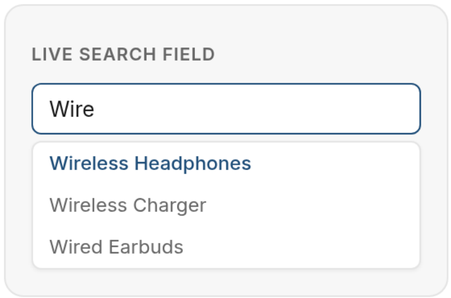


Benefit: Eliminates 80% of repetitive checkout keystrokes.

EFFORT REDUCTION 02

Predictive Text

Suggest matching terms and queries in real time as the user types.

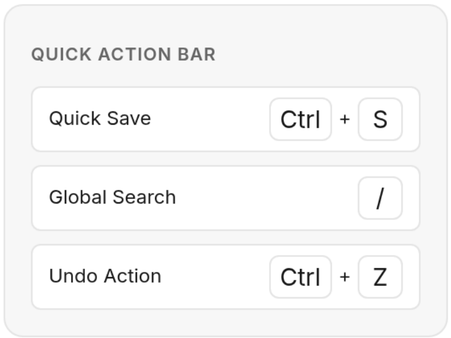


Benefit: Prevents spelling errors and narrows results instantly.

EFFORT REDUCTION 03

Shortcuts

Keyboard accelerators for high-frequency actions and power users.

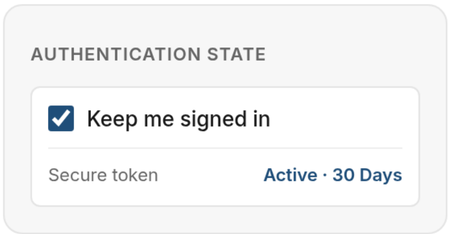


Benefit: Bypasses mouse travel for repetitive daily workflows.

EFFORT REDUCTION 04

Remember Me

Persist session tokens and user state across multiple visits.



Benefit: Eliminates continuous re-authentication friction.

3×

The Three-Clicks Rule

Users should be able to navigate to any function, document, or piece of information in no more than three mouse clicks or taps from the start screen.

Standard HCI Benchmark

UI Design Process Overview

A cyclical five-stage engineering process, with prototyping and evaluation repeated until usability criteria are met.



Scenario Development

Capture real-world user goals, workflows, and task sequences

Structure Design

Define screens and navigational relationships with an ISD

Standards Design

Establish interface metaphors, terminology, icons, and templates

Prototyping

Build iterative mockups ranging from paper sketches to code

Evaluation

Conduct heuristic reviews and user tests to validate the experience

Scenario Development & Structure Design

How task-based use scenarios directly translate into the system’s structural screen hierarchy and navigation flow.

STEP 01 · TASK ANALYSIS — SCENARIOS

Use Scenario Development

Outlines the exact steps a user follows to complete a specific goal.

E-COMMERCE SCENARIO — **Purchasing Album**

1. User searches by artist name
2. Auditions track sample & selects format
3. Adds to cart & proceeds to checkout
4. Confirms shipping address & submits payment

User Goals

Clear target outcome driving the session.

Workflows

Sequential path from entry point to completion.

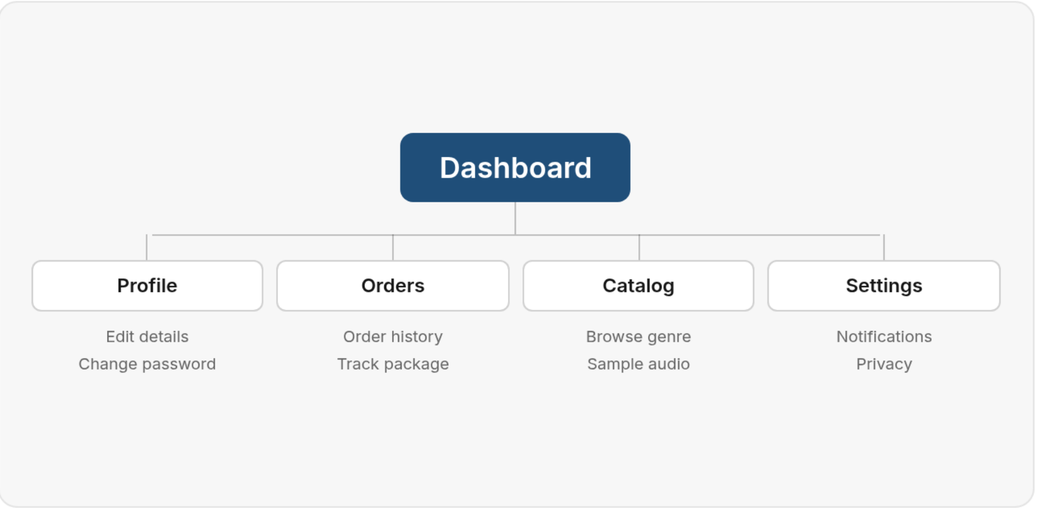
Edge Cases & Errors

Managing stockouts or payment declines gracefully.

STEP 02 · ARCHITECTURAL MAP — STRUCTURE (ISD)

Interface Structure Design

An Interface Structure Diagram (ISD) maps every screen and navigational route.



Key Integration: Scenarios identify what screens are needed; the ISD defines how users transition between them without getting trapped.

Standards Design & Prototyping

Interface standards establish design conventions, while iterative prototyping moves concepts from sketch to functional code.

INTERFACE STANDARDS

Agreed-upon design rules and conventions shared by every screen in the system.

Interface Metaphors

Real-world concepts guiding user intuition — shopping cart, desktop, folder, media player.

Objects, Actions & Icons

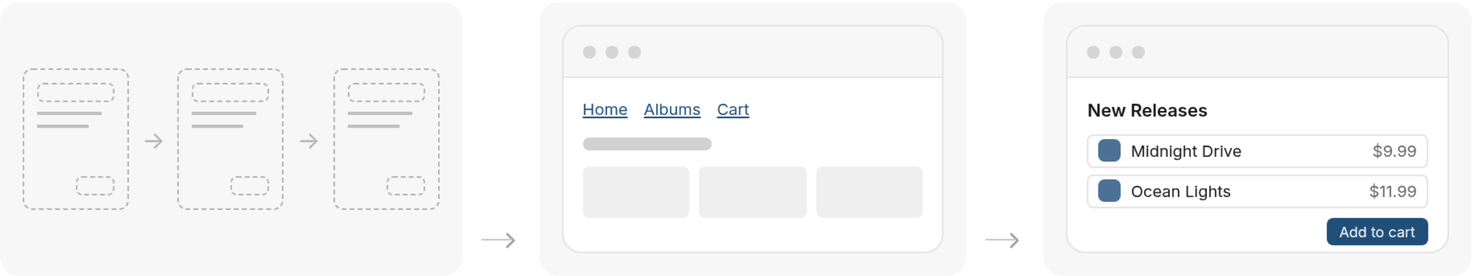
Standardized terminology and iconography across every module (e.g., “Archive” vs “Delete”).

Screen Templates

Common grid layouts and navigational zones reused to enforce structural consistency.

INTERFACE PROTOTYPING

Progressive mockups that validate visual layout, interaction flows, and user requirements.



Storyboards

Paper sketches of screens and transitions — rapid and cheap

HTML Prototypes

Clickable web wireframes with active links but static data

Language Prototypes

Built in target programming language — near-final realism

Low fidelity · quick, low cost, disposable — High fidelity · realistic, measurable, near-production

Interface Evaluation

Four evaluation methodologies with progressively deeper user involvement, realism, and analytical rigor.

01

Heuristic Evaluation

Usability experts inspect the interface against established design heuristics (e.g. Nielsen's principles).

USER INVOLVEMENT

None — experts only

Strength: Catches 70%+ of major design flaws early and cheaply

02

Walk-through Evaluation

The project team walks users through the prototype, explaining workflows and gathering feedback.

USER INVOLVEMENT

Users watch & discuss

Strength: Validates conceptual understanding without working code

03

Interactive Evaluation

Users work directly with the prototype on assigned tasks while an analyst observes hesitations and errors.

USER INVOLVEMENT

Users try hands-on

Strength: Reveals unexpected usability bottlenecks in real time

04

Formal Usability Testing

Controlled laboratory experiment recording time-on-task, completion rates, and user interactions.

USER INVOLVEMENT

Users measured rigorously

Strength: Generates objective, quantitative usability benchmark data

Quicker · lower cost · early cycle → More realistic · empirical metrics · high investment

Navigation Design

Guiding principles and primary control mechanisms that ensure seamless, reversible user movement.

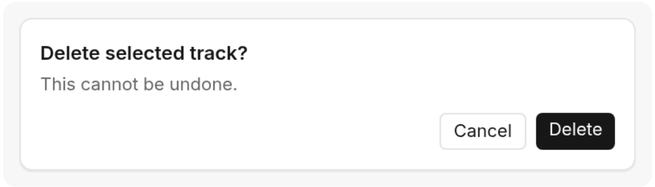
CORE NAVIGATION PRINCIPLES

Safety & Predictability

01 — ERROR DEFENSE

Prevent Mistakes

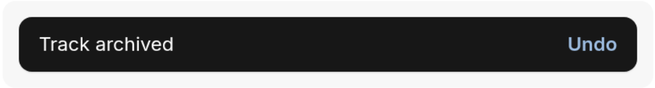
Label commands clearly, disable invalid choices, and require confirmation for destructive actions.



02 — RESILIENCE

Simplify Recovery

Ensure actions are easily reversible; keep Undo, Cancel, and Back options visible and immediate.



03 — CONSISTENCY

Consistent Grammar

Standardize command syntax uniformly: either always object-then-action or action-then-object.



NAVIGATION CONTROLS

Interaction Paradigms

Language Commands

Users type exact textual commands or natural syntax in a command interface.

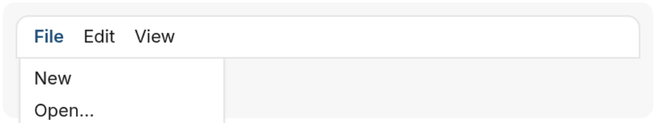


Pro: Fast for experts

Con: Requires recall

Menus

Users choose from a categorized visual list of available options and functions.

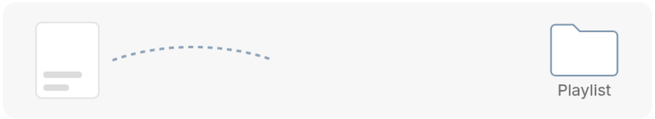


Pro: Zero memorization

Con: Slower for power users

Direct Manipulation

Users interact with visual objects directly via drag-and-drop, clicks, or gestures.



Pro: Natural tactile feedback

Con: Hard for bulk tasks

Navigation Components & Menu Types

Comparative taxonomy of all major navigation controls, component structures, and ideal deployment scenarios.

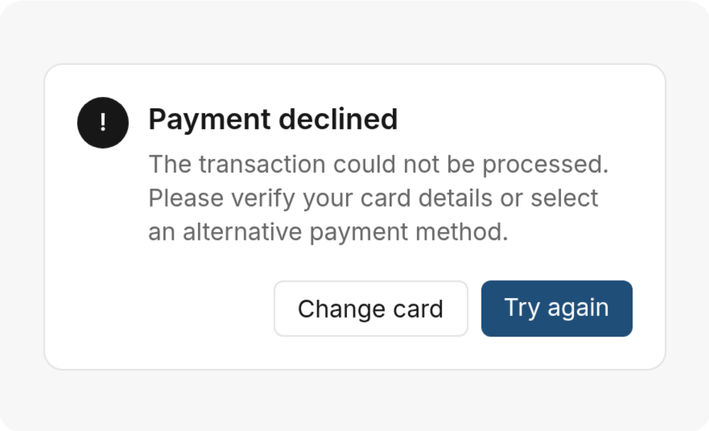
COMPONENT / CONTROL	CATEGORY	STRUCTURAL MECHANISM	BEST USE CASE / RECOMMENDATION
Menu Bar	Global Nav	Persistent horizontal list of commands across the top of the application	Primary top-level entryway for core system modules and major functional areas
Drop-down Menu	Hierarchical	Cascading menu list that drops below a parent item upon hover or click	Grouping second-level commands beneath broader menu-bar categories without clutter
Hyperlink Menu	Page Index	Structured list of text links leading to distinct pages or system sections	Main navigational index of web applications, sidebars, and resource directories
Embedded Hyperlink	In-Content	Clickable links embedded directly within paragraph copy and descriptive text	Referencing definitions, related documentation, or cross-cutting items in context
Pop-up / Context Menu	Contextual	Floats at current cursor coordinates upon right-click or long-press	Power-user shortcuts tailored specifically to the currently selected object
Tab Menu / Tabs	In-Page Views	Notebook-style tab headers switching between parallel views on the same screen	Organizing multi-panel forms, settings, or related views without navigating away
Breadcrumbs	Hierarchical Path	Linear trail showing parent-child hierarchy (Home › Music › Rock › Album)	Deep taxonomies where users must see their location and easily step backward
Search Bar	Direct Access	Text input with live query matching, auto-suggestions, and shortcut accelerator	Expansive systems where users know target keywords; bypasses deep browsing

Selection Rule: Match component visibility to usage frequency — global structures stay pinned, contextual controls appear on demand.

10 Navigation Controls Compared

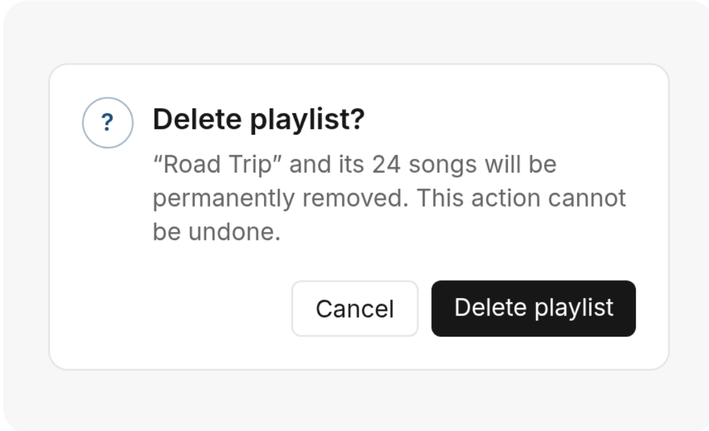
Message Design

Effective system feedback guides user action, prevents data loss, and preserves user confidence.



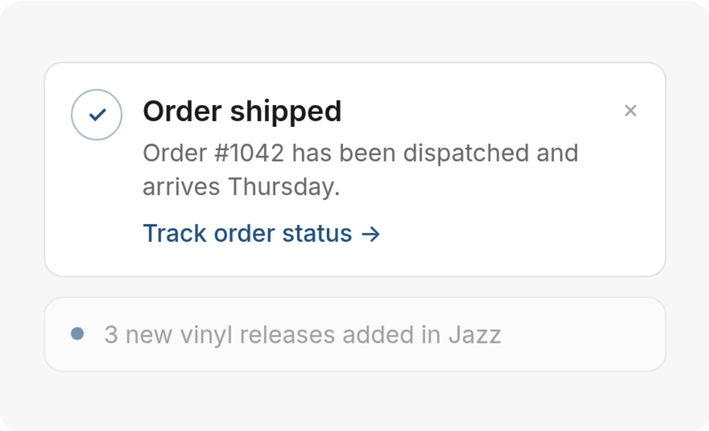
Error Messages

State clearly what happened and how to fix it in constructive terms — never blame the user or display cryptic machine codes.



Confirmation Messages

Prompt only before irreversible or destructive actions; explicitly name the action on the confirmation button.



Notification Messages

Acknowledge system state changes or background completions non-intrusively without blocking active work.

GOLDEN RULES Be clear, concise, polite, and actionable — speak in user-oriented vocabulary, never raw system errors.

HCI Feedback Standard

Input Design Overview

Capturing accurate information at the source through human-centered design principles and automated hardware capture.

CORE INPUT DESIGN PRINCIPLES

Accuracy & Integrity

01

Data Accuracy

Capture error-free data at the point of origination to prevent cascading corruption.

02

Data Integrity

Enforce business rules, schema constraints, and valid relational references.

03

Keystroke Reduction

Minimise manual typing with defaults, structured selection lists, and auto-complete.

04

Source Data Automation

Capture data electronically where it originates to eliminate human re-keying.

DATA CAPTURE TECHNOLOGIES (SOURCE DATA AUTOMATION)

Hardware & Electronic Entry

Barcode Reader

Scans 1D/2D optical codes instantly

Used in: Retail POS, warehouse inventory

OCR

Converts printed/written characters to digital data

Used in: Forms, cheques, paper archives

Smart Card

Cryptographic microchip stores and authenticates data

Used in: EMV bank cards, national IDs

RFID

Radio tags read wirelessly without line of sight

Used in: Toll systems, supply chain logistics

Web Forms

Captures structured user input directly online

Used in: E-commerce, customer portals

Online vs Batch Processing

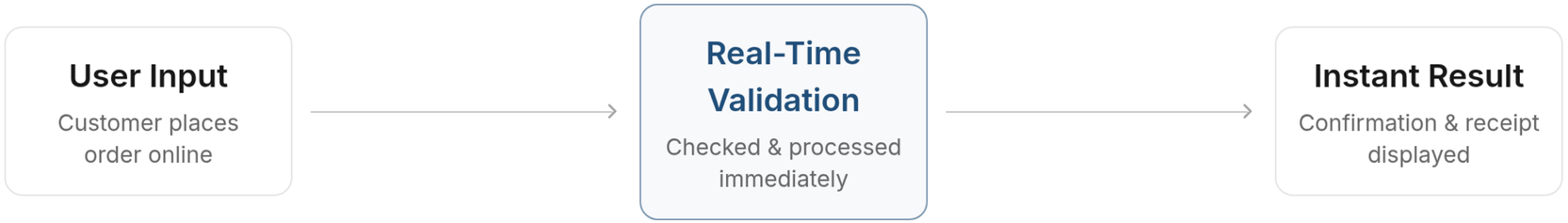
Architectural comparison between real-time transaction handling and queued batch execution.

Online Processing

Immediate

Each transaction is processed, validated, and persisted the moment it is submitted.

e.g. ATM cash withdrawal, flight seat reservation, e-commerce checkout

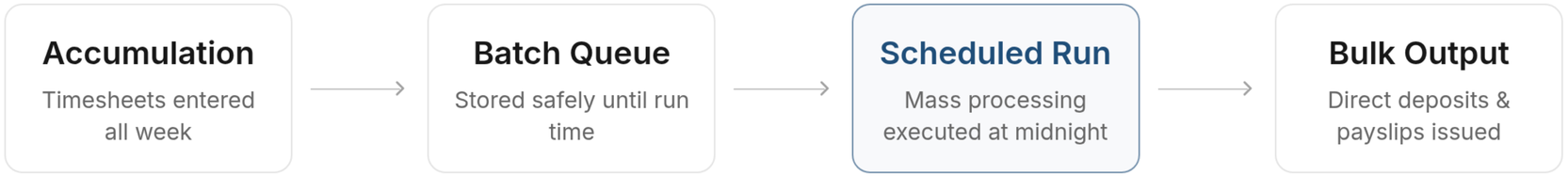


Batch Processing

Scheduled

Transactions are gathered over an operational period and processed together at scheduled times.

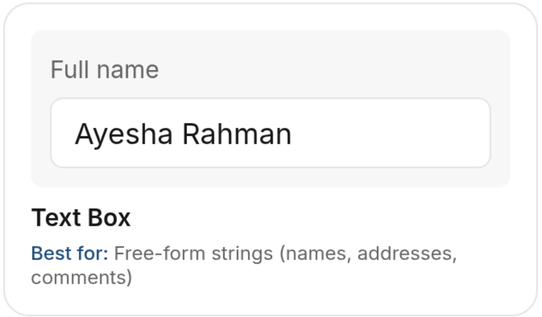
e.g. Corporate payroll, monthly utility billing, end-of-day bank clearing



Core Design Recommendation: Prefer online processing whenever possible because errors are flagged instantly while the user is actively present to rectify them.

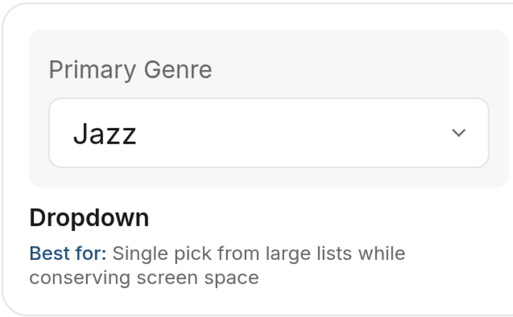
Input Controls

Selecting the optimal form control to minimize typing, enforce data integrity, and accelerate user entry.



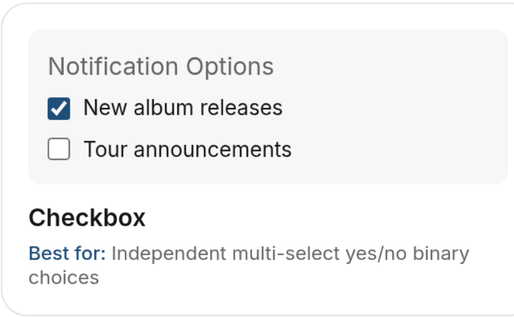
Text Box

Best for: Free-form strings (names, addresses, comments)



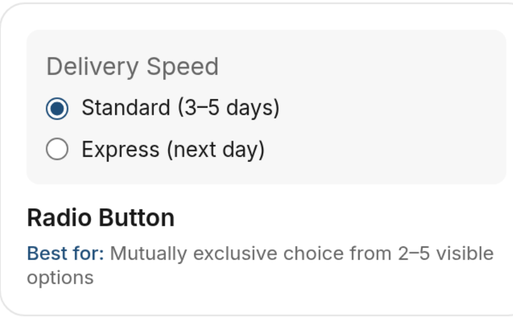
Dropdown

Best for: Single pick from large lists while conserving screen space



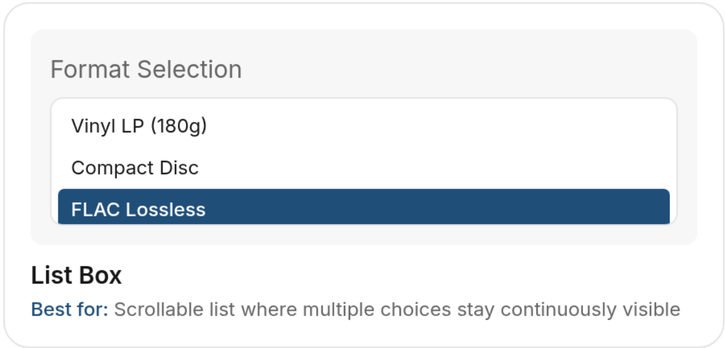
Checkbox

Best for: Independent multi-select yes/no binary choices



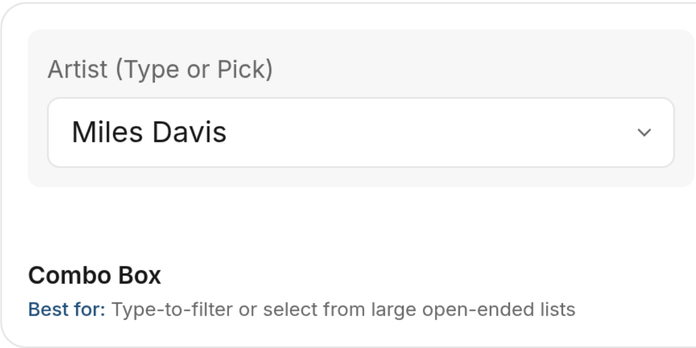
Radio Button

Best for: Mutually exclusive choice from 2–5 visible options



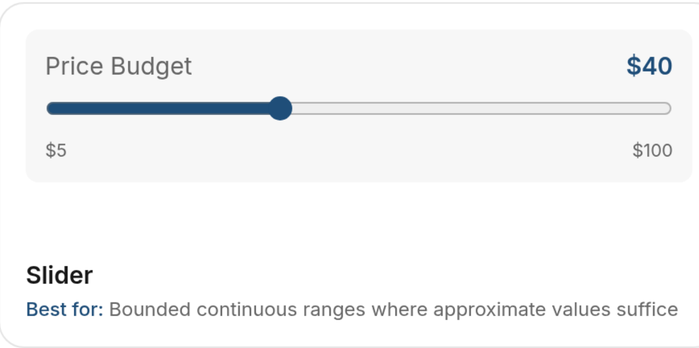
List Box

Best for: Scrollable list where multiple choices stay continuously visible



Combo Box

Best for: Type-to-filter or select from large open-ended lists



Slider

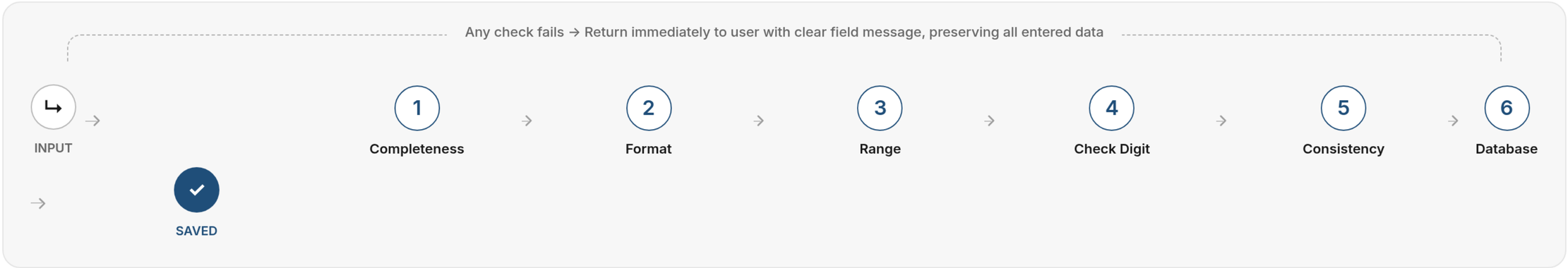
Best for: Bounded continuous ranges where approximate values suffice

Input Validation

Multi-tier validation pipeline ensuring data accuracy, integrity, and safety before database persistence.

VALIDATION PIPELINE

Sequential Verification Stages



THE SIX ESSENTIAL VALIDATION CHECKS

Comprehensive Audit Criteria

Check 1
Completeness Check
All required fields present

"Full Name is required"

Check 2
Format Check
Matches expected type & syntax

"Must be a valid email (@)"

Check 3
Range Check
Values within logical bounds

"Quantity must be 1–10"

Check 4
Check Digit Check
Mathematical algorithm check

"Luhn checksum verified"

Check 5
Consistency Check
Interdependent fields agree

"End date after start date"

Check 6
Database Check
Referential record integrity

"SKU in inventory & in stock"

Output Design & Tune Source Case Study

Foundational output principles synthesized with an end-to-end customer journey demonstrating how all lecture domains unite.

OUTPUT DESIGN FOUNDATIONS

Principles · Types · Formats

Output Principles

Core Quality

Relevance
Present only information necessary for user tasks and decisions

Accuracy
Verifiable, error-free data free from presentation bias

Timeliness
Delivered when needed so users can take immediate action

Accessibility
Readable across devices, contrast requirements, and abilities

Output Types

Presentation

Reports: Structured documents: detail, summary, exception, and turnaround records

Dashboards: At-a-glance visualization of operational KPIs and live audio streams

Notifications: Actionable alerts delivered via push, SMS, email, and toasts

Output Formats

Delivery Media

Print	Web	Mobile	PDF	Excel
Formal archival records	Interactive dynamic views	On-the-go notifications	Fixed-layout invoices	Analytical exports

TUNE SOURCE CASE STUDY · END-TO-END USER JOURNEY

How All 5 Disciplines Work Together

1 → 2 → 3 → 4 → 5

UI & NAVIGATION
Browse Music
Filter by genre or search by artist

System: Catalog grid, persistent search bar, and genre breadcrumbs.

UI & INPUT CONTROLS
View Album
Audition track sample, select media format

System: Audio waveform player, format dropdown (Vinyl/CD/FLAC).

NAVIGATION & FEEDBACK
Add To Cart
Verify quantity & shipping choice

System: Interactive badge update, non-blocking toast confirmation.

INPUT & VALIDATION
Checkout
Enter address, shipping, and payment

System: Autofill address; inline format, range & Luhn card validation.

OUTPUT DESIGN
Confirmation
Review invoice, download files, tracking

System: Instant dashboard receipt, PDF invoice, digital vault token.

Architectural Synergy: UI Design attracts & orientates → Navigation guides → Input & Validation capture error-free data → Output confirms & delivers value.

Tune Source Music Store System

Interactive Seminar Activity

Q SEMINAR DISCUSSION PROMPT

How does consistency improve user experience?

Analyze the compounding impact of internal and external consistency on user confidence, learning curves, and system errors.

Cognitive Load Reduction
Users transfer existing mental models effortlessly without having to pause, decipher, or relearn interactions.

Error Prevention & Safety
Standardized button positions, destructive confirmations, and color semantics prevent accidental mistakes.

Speed & Muscle Memory
Predictable workflows and keyboard accelerators allow frequent users to navigate at maximum throughput.

Design System Governance
Centralized UI tokens, reusable component templates, and heuristic audits ensure coherence across teams.

LECTURE SUMMARY · CORE CURRICULUM PILLARS

Complete Topic Synthesis

01 PILLAR 1
UI Principles
Balance layout, aesthetics, UX, consistency, and effort minimization (the Three-Clicks Rule).

02 PILLAR 2
UI Process
Iterate systematically from use scenarios through structure to prototyping and evaluation.

03 PILLAR 3
Navigation Design
Provide predictable hierarchies, mistake prevention, easy recovery, and appropriate menu controls.

04 PILLAR 4
Input Design
Capture data at source via automation, select the optimal control, and minimize manual typing.

05 PILLAR 5
Validation
Enforce multi-tier checks (completeness, format, range, check digit, consistency, database).

06 PILLAR 6
Output Design
Deliver relevant, accurate, timely reports and dashboards across web, mobile, PDF, and print.

Thank You.

Questions & Academic Discussion

System Analysis & Design
Department of Computer Science & Engineering
● Course Lecture Presentation Complete · 20 Slides